AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

FINAL ASSESSMENT TEST

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QUESTION: (SET – 6)

Smart Electricity Billing System

Problem Statement:

Ensure electricity usage is calculated and billed correctly.

Tasks:

1. Develop billing calculation logic

2. Write JUnit tests for tariff calculations

3. Implement CI/CD pipeline with automated testing

➤Set up version control using GitHub

Configure Cl pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits

Automate build process using Maven

Integrate automated unit tests in pipeline

➤ Configure CD pipeline for automatic deployment to staging/production

4. Deploy services on Docker and Kubernetes

Containerize application using Docker (write Dockerfile)

Build and push Docker image to container registry (Docker Hub)

Create Kubernetes deployment and service YAML files

OUTPUT

FILE STRUCTURE

EXPLORER

SMART-ELECTRICITY-BILLING

- .github\workflows
  - ci.yml M
- src
  - main\java\com\example\billing
    - BillingService.java 1
    - Main.java 1
  - test\java\com\example\billing
    - BillingServiceTest.java 9+
- Dockerfile M
- pom.xml M
- kubernetes-deployment.yaml M
- kubernetes-service.yaml M

kubernetes-deployment.yaml M | kubernetes-service.yaml M | ci.yml M | pom.xml M

.github > workflows > ci.yml

```yaml
name: Java CI Pipeline

on:
  push:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    - name: Checkout Code
      uses: actions/checkout@v3

    - name: Set up JDK
      uses: actions/setup-java@v3
      with:
        java-version: '17'
```

- src
  - main\java
    - App.java
  - test\java
    - TestApp.java
- target
- deployment.yaml
- Dockerfile
- pom.xml
- service.yaml

OUTLINE
TIMELINE
JAVA PROJECTS

```
[INFO] Building library-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ library-system ---
[INFO] Deleting C:\Users\Suresh\OneDrive\Documents\FAT\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ library-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\FAT\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ library-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ library-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\FAT\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ library-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ library-system ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.932 s
[INFO] Finished at: 2026-04-13T10:03:08+05:30
```

billing Public

Pin | Watch 0 | Fork 0 | Star 0

Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
Get started with GitHub Copilot

Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators

Quick setup — if you've done this kind of thing before

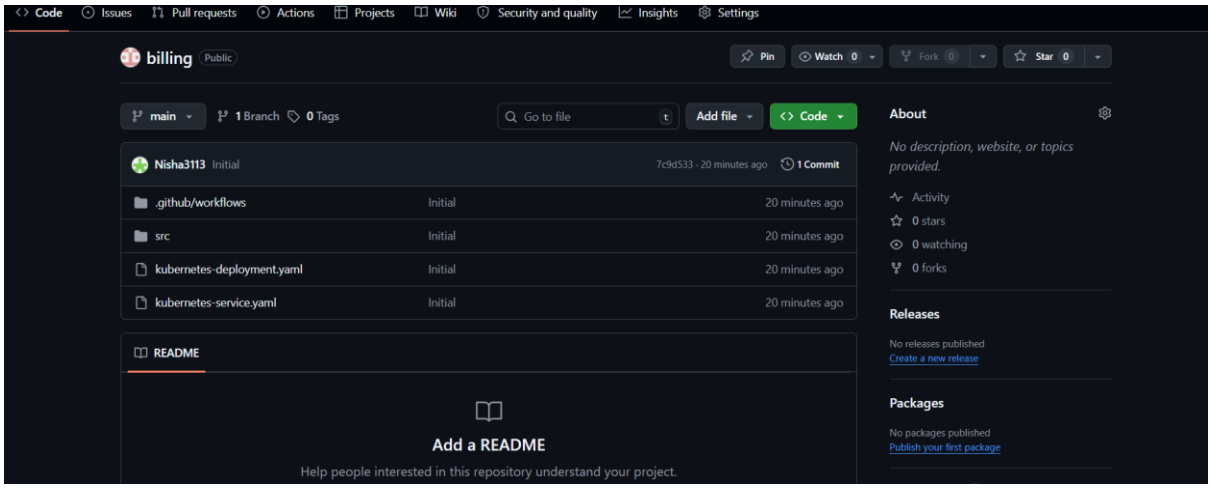
Set up in Desktop or HTTPS SSH https://github.com/abishar605/billing.git

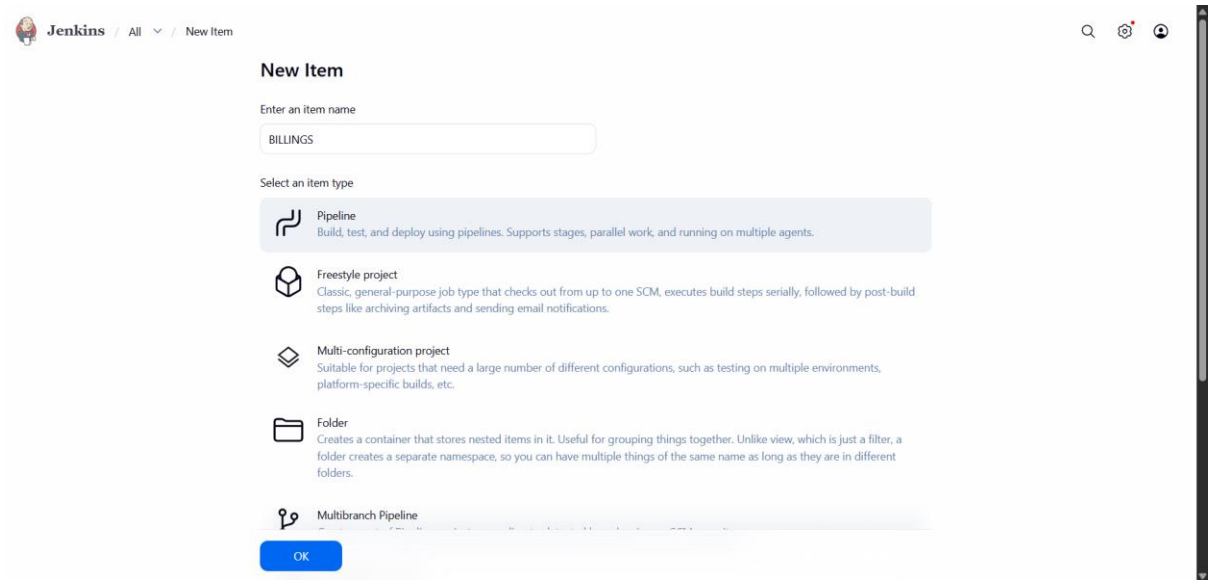
Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.

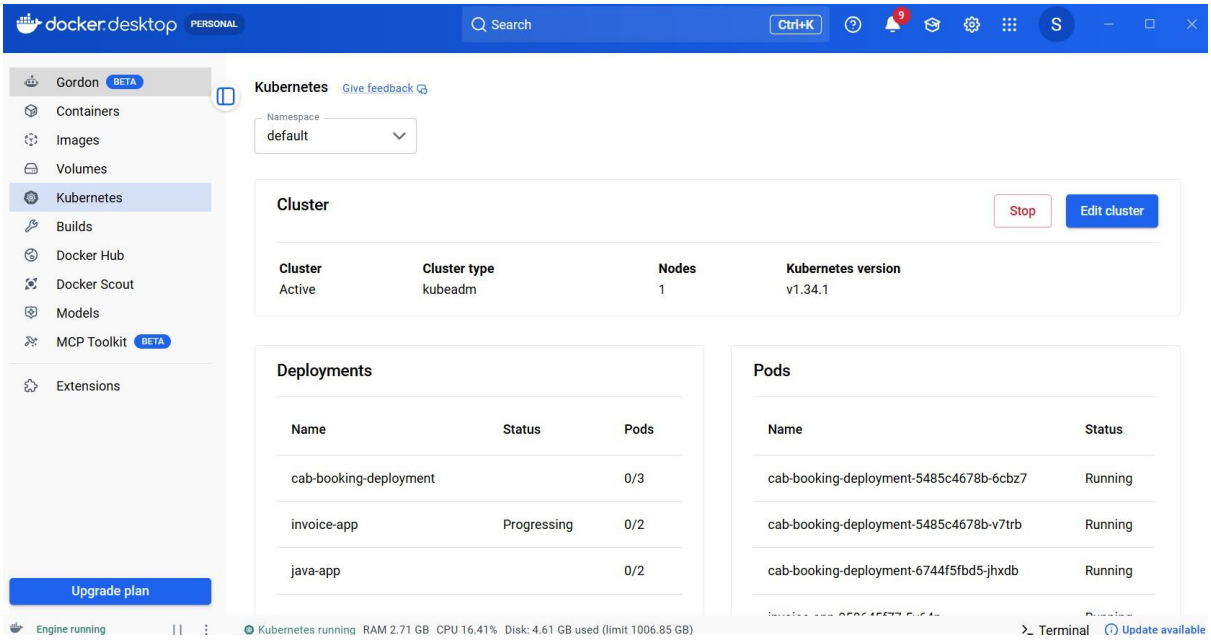
...or create a new repository on the command line

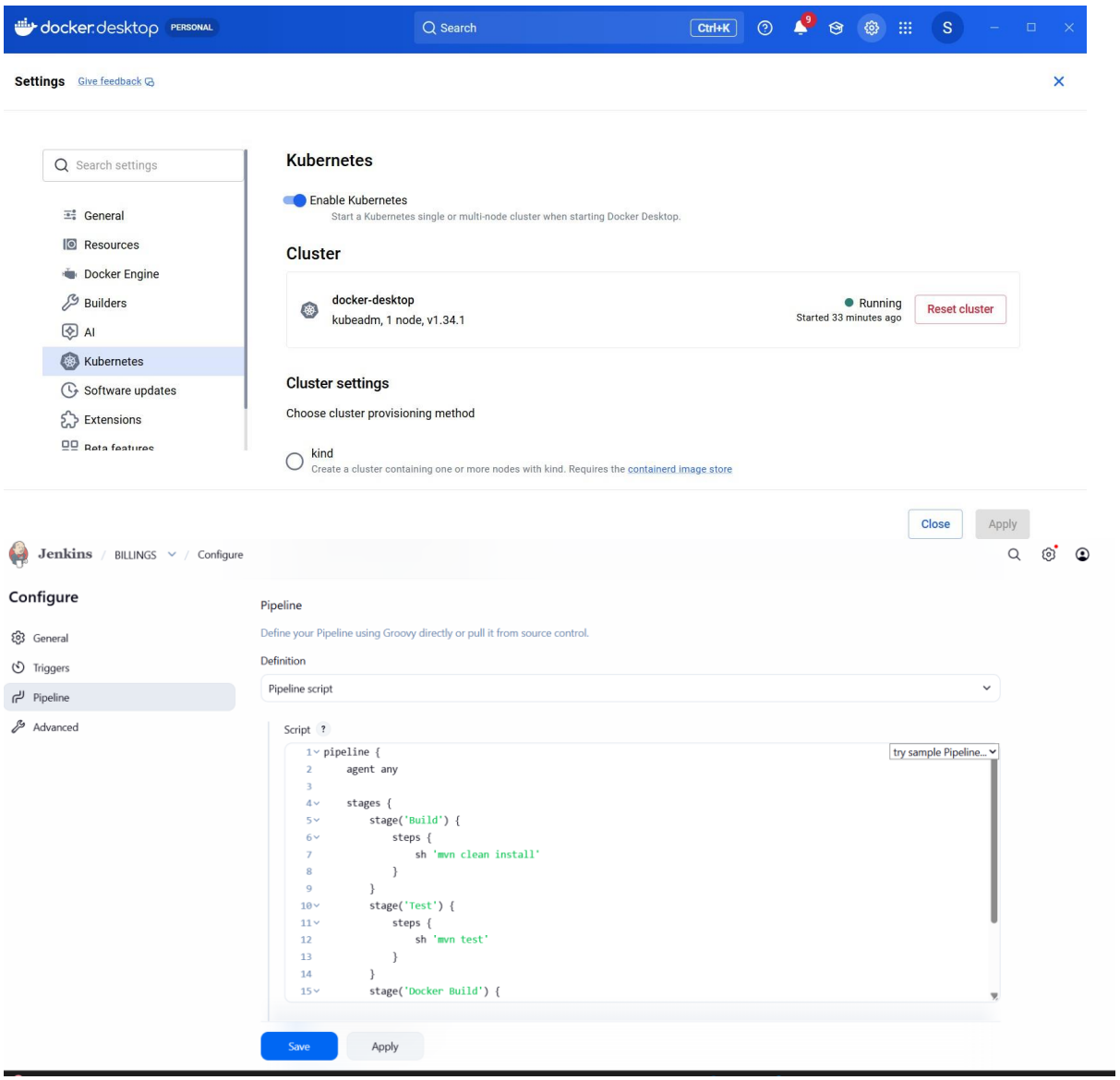
```
echo "# billing" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/abishar605/billing.git
git push -u origin main
```

```
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git init
Initialized empty Git repository in C:/Users/abish/OneDrive/Desktop/smart-electricity-billing/.git/
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git add .
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git commit -m "Initial"
[main (root-commit) 7c9d533] Initial
 8 files changed, 80 insertions(+)
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 kubernetes-deployment.yaml
 create mode 100644 kubernetes-service.yaml
 create mode 100644 src/Dockerfile
 create mode 100644 src/main/java/com/example/billing/BillingService.java
 create mode 100644 src/main/java/com/example/billing/Main.java
 create mode 100644 src/pom.xml
 create mode 100644 src/test/java/com/example/billing/BillingServiceTest.java
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git branch -M main
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git remote add origin https://github.com/abishar605
/billing.git
PS C:\Users\abish\OneDrive\Desktop\smart-electricity-billing> git push -u origin main
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (20/20), 1.90 KiB | 108.00 KiB/s, done.
Total 20 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/abishar605/billing.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```






docker desktop PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kubeadm
Nodes
1
Kubernetes version
v1.34.1
Deployments
Name
Status
Pods
cab-booking-deployment
0/3
invoice-app
Progressing
0/2
java-app
0/2
Pods
Name
Status
cab-booking-deployment-5485c4678b-6cbz7
Running
cab-booking-deployment-5485c4678b-v7trb
Running
cab-booking-deployment-6744f5fbd5-jhxdb
Running
Engine running
Kubernetes running RAM 2.71 GB CPU 16.41% Disk: 4.61 GB used (limit 1006.85 GB)
Terminal
Update available


docker desktop PERSONAL
Search
Ctrl+K
Settings
Give feedback
Search settings
General
Resources
Docker Engine
Builders
AI
Kubernetes
Software updates
Extensions
Beta features
Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
Cluster
docker-desktop
kubeadm, 1 node, v1.34.1
Running
Started 33 minutes ago
Reset cluster
Cluster settings
Choose cluster provisioning method
kind
Create a cluster containing one or more nodes with kind. Requires the containerd image store
Close
Apply
Jenkins / BILLINGS / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
try sample Pipeline...
pipeline {
agent any
stages {
stage('Build') {
steps {
sh 'mvn clean install'
}
}
stage('Test') {
steps {
sh 'mvn test'
}
}
stage('Docker Build') {
Save
Apply

```
C:\ProgramData\Jenkins\.jenkins\workspace\Library-Book-Management-CI-CD>echo Services:
Services:

C:\ProgramData\Jenkins\.jenkins\workspace\Library-Book-Management-CI-CD>kubectl get svc
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
cab-booking-service NodePort    10.100.204.129   <none>        80:30009/TCP   3h21m
invoice-service     NodePort    10.101.108.45    <none>        80:30007/TCP   5d21h
java-service        NodePort    10.106.182.87    <none>        80:30008/TCP   7d1h
kubernetes          ClusterIP   10.96.0.1        <none>        443/TCP        7d1h
library-service     NodePort    10.109.15.251    <none>        80:32603/TCP   3s
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
✅ Library System CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

BillingService.java

```java
package com.example.billing;

public class BillingService {

    public double calculateBill(int units) {
        double bill = 0;

        if (units <= 100) {
            bill = units * 1.5;
```

```
        } else if (units <= 200) {
            bill = (100 * 1.5) + (units - 100) * 2.5;
        } else {
            bill = (100 * 1.5) + (100 * 2.5) + (units - 200) * 4;
        }

        return bill;
    }
}
```

Main.java

```
package com.example.billing;

public class Main {
    public static void main(String[] args) {
        BillingService service = new BillingService();
        int units = 250;

        double bill = service.calculateBill(units);
        System.out.println("Electricity Bill: ₹" + bill);
    }
}
```

BillingServiceTest.java

```
package com.example.billing;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class BillingServiceTest {

    BillingService service = new BillingService();

    @Test
    void testBelow100Units() {
        assertEquals(150, service.calculateBill(100));
    }

    @Test
    void testBetween100And200Units() {
        assertEquals(275, service.calculateBill(150));
    }

    @Test
    void testAbove200Units() {
```

```
        assertEquals(650, service.calculateBill(250));
    }
}
```

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>billing</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

    <dependencies>
        <!-- JUnit 5 -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.9.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <!-- Surefire for running tests -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0-M5</version>
            </plugin>
        </plugins>
    </build>

</project>
```

Dockerfile

```
FROM openjdk:17
COPY target/billing-1.0-SNAPSHOT.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

ci.yml

```yaml
name: Java CI Pipeline

on:
  push:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    - name: Checkout Code
      uses: actions/checkout@v3

    - name: Set up JDK
      uses: actions/setup-java@v3
      with:
        java-version: '17'

    - name: Build with Maven
      run: mvn clean install

    - name: Run Tests
      run: mvn test
```

Kubernetes-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: billing-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: billing
  template:
    metadata:
      labels:
        app: billing
    spec:
```

```yaml
      containers:
      - name: billing-container
        image: your-dockerhub-username/billing-app
        ports:
        - containerPort: 8080
```

Kubernetes-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: billing-service
spec:
  type: NodePort
  selector:
    app: billing
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30007
```